

David Ginsburger's notes, 2021

Lecture notes: detour on RKHSs

Let us consider a set D and denote by $\mathbb{R}^D$ the set of functions from D to $\mathbb{R}$. For any $\mathbf{x} \in D$, let us denote by $e_{\mathbf{x}}$ the evaluation functional at $\mathbf{x}$, that is, the mapping defined by $e_{\mathbf{x}} : f \in \mathbb{R}^D \rightarrow e_{\mathbf{x}}(f) = f(\mathbf{x}) \in \mathbb{R}$. One can observe that, by construction, such a mapping is linear on $\mathbb{R}^D$, as for any $f, g \in \mathbb{R}^D$ and $\alpha \in \mathbb{R}$, $e_{\mathbf{x}}(f + \alpha g) = (f + \alpha g)(\mathbf{x}) = f(\mathbf{x}) + \alpha g(\mathbf{x}) = e_{\mathbf{x}}(f) + \alpha e_{\mathbf{x}}(g)$. Evaluation functionals play a central role in the theory of Reproducing Kernel Hilbert Spaces (RKHSs) as detailed below. Let us define for a start what we mean by reproducing kernels and associated RKHSs.

Definition 1. A Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ of functions $D \rightarrow \mathbb{R}$ is said to be a Reproducing Kernel Hilbert Space if there exists a function $k : D \times D \rightarrow \mathbb{R}$ such that

- $\forall \mathbf{x} \in D, k(\mathbf{x}, \cdot) \in \mathcal{H}$, and
- $\forall \mathbf{x} \in D, \forall f \in \mathcal{H}, f(\mathbf{x}) = \langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}}$.

Such a function k is then referred to as reproducing kernel of $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$.

Unicity of the reproducing kernel of a RKHS follows directly from this definition. Assuming indeed that a RKHS $\mathcal{H}$ (the scalar product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ of which being sometimes not explicited for brevity) possesses two reproducing kernels k_1, k_2 , we would then have for any $\mathbf{x}, \mathbf{x}' \in D$,

$$\begin{aligned} k_1(\mathbf{x}, \mathbf{x}') &= \langle k_1(\mathbf{x}, \cdot), k_2(\mathbf{x}', \cdot) \rangle_{\mathcal{H}} \quad (\text{by Def. 1, applied to } k_2) \\ &= \langle k_2(\mathbf{x}', \cdot), k_1(\mathbf{x}, \cdot) \rangle_{\mathcal{H}} \quad (\text{by inner product symmetry}) \\ &= \langle k_2(\mathbf{x}', \cdot), k_2(\mathbf{x}, \cdot) \rangle_{\mathcal{H}} \quad (\text{by Def. 1, applied to } k_2) \\ &= k_2(\mathbf{x}, \mathbf{x}') \quad (\text{by inner product symmetry and Def. 1, applied to } k_2). \end{aligned}$$

One can hence speak of the reproducing kernel k of $\mathcal{H}$. This particular name is often informally attributed in the literature to the fact that the inner product of the kernel with itself (at two respective points, in the sense of the first line in the equation above) delivers the kernel again (at the considered pair of points). Another common interpretation of this name is that taking the inner product of f with $k(\mathbf{x}, \cdot)$ "reproduces" the value of f at $\mathbf{x}$. In the following, we will typically use the notation $k_{\mathbf{x}}$ to denote the function $k(\mathbf{x}, \cdot)$, leading to a more concise formulation of the reproducing property given by $k(\mathbf{x}, \mathbf{x}') = \langle k_{\mathbf{x}}, k_{\mathbf{x}'} \rangle_{\mathcal{H}}$ (resp. $f(\mathbf{x}) = \langle f, k_{\mathbf{x}} \rangle_{\mathcal{H}}$).

We now come back to evaluation functionals, and we establish that their continuity on a Hilbert space of functions actually constitutes a characterization of RKHSs.

Property 1. A Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ of functions $D \rightarrow \mathbb{R}$ is a Reproducing Kernel Hilbert Space if and only if, for any $\mathbf{x} \in D$, $e_{\mathbf{x}}$ is continuous on $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$. $e_{\mathbf{x}} : f \mapsto f(\mathbf{x})$

Proof. Assuming that $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ is a RKHS with reproducing kernel k , Def. 1 and an application of the Cauchy-Schwarz inequality imply that for any $\mathbf{x} \in D, f \in \mathcal{H}$,

$$|e_{\mathbf{x}}(f)| = |\langle f, k_{\mathbf{x}} \rangle_{\mathcal{H}}| \leq \|f\|_{\mathcal{H}} \|k_{\mathbf{x}}\|_{\mathcal{H}},$$

À tout espace de Hilbert, il existe un noyau reproduisant (si $\forall \mathbf{x}, e_{\mathbf{x}}$ est continue)

so that $e_{\mathbf{x}}$ is indeed a continuous —or “bounded”— functional (for any sequence $(f_n)_{n \geq 1}$ in $\mathcal{H}$ such that $(\|f_n\|_{\mathcal{H}})_{n \geq 1}$ converges to 0, $(e_{\mathbf{x}}(f_n))_{n \geq 1}$ converges to 0).

Reciprocally, if a Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ of functions $D \rightarrow \mathbb{R}$ is such that the evaluation functionals are continuous on $\mathcal{H}$ (with respect to the norm associated with its inner product), then, for any $\mathbf{x} \in D$, we can apply Riesz’s “representer theorem” guaranteeing the existence of a “representer” $R_{\mathbf{x}} \in \mathcal{H}$ for $e_{\mathbf{x}}$, i.e.

$$\forall f \in \mathcal{H}, e_{\mathbf{x}}(f) = \langle f, R_{\mathbf{x}} \rangle_{\mathcal{H}}.$$

Defining $k : D \times D \rightarrow \mathbb{R}$ by $k(\mathbf{x}, \mathbf{x}') = R_{\mathbf{x}}(\mathbf{x}')$, we then have on the one hand that for any $\mathbf{x} \in D$, $k(\mathbf{x}, \cdot) = R_{\mathbf{x}}(\cdot) \in \mathcal{H}$ and, on the other hand, that for any $\mathbf{x} \in D$ and $f \in \mathcal{H}$, $\langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}} = \langle f, R_{\mathbf{x}} \rangle_{\mathcal{H}} = f(\mathbf{x})$, so that k qualifies as $\mathcal{H}$ ’s kernel. $\square$

Corollary 1. *Let $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ be a RKHS and $(f_n)_{n \geq 1}$ be a sequence in $\mathcal{H}$ converging to $f \in \mathcal{H}$ in the RKHS norm. Then, for any $\mathbf{x} \in D$, $(f_n(\mathbf{x}))_{n \geq 1}$ converges to $f(\mathbf{x})$.*

Proof. This directly follows from the representation of $e_{\mathbf{x}}$ by $k_{\mathbf{x}}$, since for all $n \geq 1$,

$$|f_n(\mathbf{x}) - f(\mathbf{x})| = |e_{\mathbf{x}}(f_n) - e_{\mathbf{x}}(f)| = |\langle f_n - f, k_{\mathbf{x}} \rangle_{\mathcal{H}}| \leq \|f_n - f\|_{\mathcal{H}} \|k_{\mathbf{x}}\|_{\mathcal{H}},$$

whereof convergence of $\|f_n - f\|_{\mathcal{H}}$ to 0 implies indeed that $f_n(\mathbf{x}) \xrightarrow[n \rightarrow \infty]{} f(\mathbf{x})$. $\square$

Before we continue our excursion into the theoretical foundations of RKHSs, let us now present and discuss some important examples and counter-examples.

Example 1. *Let us consider a finite-dimensional vector space $\mathcal{H}$ of functions $D \rightarrow \mathbb{R}$ equipped with a basis $v_1, \dots, v_n$. One can note that in such a case, any inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ on $\mathcal{H}$ is completely characterized by its values on pairs of basis elements, i.e. by the matrix $G = (\langle v_i, v_j \rangle_{\mathcal{H}})_{i,j \in \{1, \dots, n\}}$. Indeed, for any $f, g \in \mathcal{H}$, there exist $\alpha_i, \beta_i \in \mathbb{R}$ ($1 \leq i \leq n$) such that $f = \sum_{i=1}^n \alpha_i v_i$ and $g = \sum_{i=1}^n \beta_i v_i$, whereof $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j \langle v_i, v_j \rangle_{\mathcal{H}} = \alpha^T G \beta$ (where $\alpha = (\alpha_1, \dots, \alpha_n)^T$ and similarly for β). In particular, G inherits from the inner product’s symmetry and positive definiteness. Given now an inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ on $\mathcal{H}$ associated with a given symmetric positive definite matrix G , we claim that $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ is a RKHS with reproducing kernel $k(\mathbf{x}, \mathbf{x}') = \sum_{i=1}^n \sum_{j=1}^n G^{ij} v_i(\mathbf{x}) v_j(\mathbf{x}')$, where G^{ij} stands for the (i, j) -th entry of the inverse matrix G^{-1} . Let us prove it by checking that Def. 1’s conditions are fulfilled.*

First, for any $\mathbf{x} \in D$, $k(\mathbf{x}, \cdot) = \sum_{j=1}^n \sum_{i=1}^n G^{ij} v_i(\mathbf{x}) v_j(\cdot) \in \mathcal{H}$. Second, consider an arbitrary element $f = \sum_{\ell=1}^n \alpha_{\ell} v_{\ell} \in \mathcal{H}$ and a point $\mathbf{x} \in D$. Then we have

$$\begin{aligned} \langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}} &= \left\langle \sum_{\ell=1}^n \alpha_{\ell} v_{\ell}, \sum_{i=1}^n \sum_{j=1}^n G^{ij} v_i(\mathbf{x}) v_j \right\rangle_{\mathcal{H}} \\ &= \sum_{\ell=1}^n \sum_{i=1}^n \sum_{j=1}^n G^{ij} \langle v_{\ell}, v_j \rangle_{\mathcal{H}} \alpha_{\ell} v_i(\mathbf{x}) \\ &= \sum_{i=1}^n \sum_{\ell=1}^n \delta_{i,\ell} \alpha_{\ell} v_i(\mathbf{x}) = \sum_{i=1}^n \alpha_i v_i(\mathbf{x}) = f(\mathbf{x}). \end{aligned}$$

In particular, when $v_1, \dots, v_n$ form an orthonormal basis of $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$, then the latter is a RKHS with reproducing kernel $k(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^n v_j(\mathbf{x})v_j(\mathbf{x}')$.

Notable particular cases of finite-dimensional RKHSs include for instance $\mathbb{R}^n$ spaces equipped with Euclidean inner product or variations thereof based on positive definite matrices, but also spaces of polynomials with given finite degree, or spaces spanned by $\cos(j\mathbf{x})$ and $\sin(j\mathbf{x})$ functions (where $0 \leq j \leq q$ for some given $q \geq 0$).

Example 2. We focus here on the Hilbert space $\ell^2(\mathbb{N})$ of square summable (real-valued) sequences equipped with the inner product $\langle s, s' \rangle_{\ell^2(\mathbb{N})} = \sum_{i=0}^{\infty} s_i s'_i$ ($s, s' \in \ell^2(\mathbb{N})$). This space turns out to be a RKHS, with reproducing kernel $k : (i, j) \in \mathbb{N}^2 \rightarrow k(i, j) = \delta_{i,j}$ (where $\delta_{i,j}$ denotes again the Kronecker symbol, with value 1 if $i = j$ and 0 otherwise). Indeed, for any $i \in \mathbb{N}$, $k(i, \cdot) \in \ell^2(\mathbb{N})$ as $\sum_{j=0}^{\infty} k(i, j)^2 = \sum_{j=0}^{\infty} \delta_{i,j} = 1 < \infty$, and for any $s \in \ell^2(\mathbb{N})$, $\langle s, k(i, \cdot) \rangle_{\ell^2(\mathbb{N})} = s_i$.

Example 3. Let us consider $D = [0, 1]$ equipped with the Lebesgue measure λ (restricted to D) and the space of square integrable functions $\mathcal{L}^2(\lambda) = \{f : D \rightarrow \mathbb{R} \text{ such that } \int_0^1 f(x)^2 d\lambda(x) < \infty\}$ with bilinear form $\langle f, g \rangle_{\mathcal{L}^2(\lambda)} = \int_0^1 f(x)g(x) d\lambda(x)$ ($f, g \in \mathcal{L}^2(\lambda)$). While the latter is positive semi-definite, it fails to qualify as an inner product as $\int_0^1 f(x)^2 d\lambda(x) = 0$ does not imply that f is the null function. Yet, one typically constructs from there a Hilbert space denoted $L^2(\lambda)$ by taking classes of equivalence of functions (up to a λ -almost everywhere equality). However, $L^2(\lambda)$ cannot qualify as RKHS, in the first place because it is not a function space per se (it is rather a space of classes of equivalence). Besides this, coming back for the sake of discussion to the function space $\mathcal{L}^2(\lambda)$, the continuity of evaluations is far from being met. Consider indeed the sequence $(f_n)_{n \geq 1}$ of elements of $\mathcal{L}^2(\lambda)$ defined by $f_n(x) = x^n$ ($x \in [0, 1], n \geq 1$). In the (semi-)norm associated with $\langle \cdot, \cdot \rangle_{\mathcal{L}^2(\lambda)}$, a direct calculation leads to $\|f_n\|_{\mathcal{L}^2(\lambda)}^2 = \int_0^1 x^{2n} d\lambda(x) = \frac{1}{2n+1}$ whereof $(f_n)_{n \geq 1}$ converges to 0 in the sense of this (semi-)norm. On the other hand, for any $x \in [0, 1]$ and $n \geq 1$, $f_n(1) = 1$ and so $f_n(1) \rightarrow 1$ when $n \rightarrow \infty$.

Example 4. Define $H_0^1([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R}, f \text{ AC (absolutely continuous), } f(0) = 0, f' \in L^2(\lambda)\}$, where f' refers to the weak derivative of f (i.e., an element of $L^1(\lambda)$ such that $\int_0^1 f'(x)\varphi(x) d\lambda(x) = -\int_0^1 f(x)\varphi'(x) d\lambda(x)$ for all infinitely differentiable functions $\varphi : [0, 1] \rightarrow \mathbb{R}$ with $\varphi(0) = \varphi(1) = 0$). It is known that $\langle f, g \rangle_{H_0^1} = \int_0^1 f'(x)g'(x) d\lambda(x)$ is an inner product on $H_0^1([0, 1])$ and that the resulting inner product space is complete. It turns out to be a RKHS with kernel $k(x, x') = \min(x, x')$. While, for any given $x \in [0, 1]$ the function $k_x : x' \in [0, 1] \mapsto k_x(x') \in \mathbb{R}$ presents a kink at x where it is hence not differentiable in the strong sense, it does possess a weak derivative in $L^2(\lambda)$, namely $k'_x = \mathbf{1}_{[0, x]}$. Hence for any $x \in [0, 1]$, $k_x \in H_0^1([0, 1])$, and furthermore for any $f \in H_0^1([0, 1])$ we do have

$$f(x) = \int_0^x f'(u) d\lambda(u) = \int_0^1 f'(u) \mathbf{1}_{[0, x]}(u) d\lambda(u) = \int_0^1 f'(u) k'_x(u) d\lambda(u) = \langle f, k_x \rangle_{H_0^1},$$

thereby showing that k is indeed the reproducing kernel of $H_0^1([0, 1])$. Let us remark that if the constraint that $f(0) = 0$ is removed, one can construct yet another Hilbert space by setting $H^1([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R}, f \text{ AC, } f' \in L^2(\lambda)\}$ and $\langle f, g \rangle_{H^1} =$

$f(0)g(0) + \int_0^1 f'(x)g'(x)d\lambda(x)$. It is then a routine exercise to show that this space is a RKHS as well, with reproducing kernel $k(x, x') = 1 + \min(x, x')$.

Let us now turn back to fundamental properties of RKHS and focus more precisely on the interplay between reproducing kernels and positive definiteness. Recall that we call here positive definite (p.d.) kernels symmetric functions $k : D \times D \rightarrow \mathbb{R}$ such that $\forall n \geq 1, \mathbf{x}_1, \dots, \mathbf{x}_n \in D, a_1, \dots, a_n \in \mathbb{R}, \sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0$. The first fact that we want to mention now is that if $k : D \times D \rightarrow \mathbb{R}$ is the reproducing kernel of some RKHS $\mathcal{H}$, then it is systematically p.d. First, the symmetry is directly inherited from the symmetry of the inner product. Second, considering any $n \geq 1$ and $\mathbf{x}_1, \dots, \mathbf{x}_n \in D, a_1, \dots, a_n \in \mathbb{R}$, we have by definition of a reproducing kernel:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \langle k(\mathbf{x}_i, \cdot), k(\mathbf{x}_j, \cdot) \rangle_{\mathcal{H}} \\ &= \left\langle \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot), \sum_{j=1}^n a_j k(\mathbf{x}_j, \cdot) \right\rangle_{\mathcal{H}} = \left\| \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot) \right\|_{\mathcal{H}}^2 \geq 0. \end{aligned}$$

Reciprocally, one may wonder if any p.d. kernels is the reproducing kernel of a RKHS. The answer is positive, as was established by Moore and Aronszajn (1950).

Let us give a sketch of proof by highlighting the main ideas of the underlying RKHS construction while glossing over some more technical details of it.

First, consider the space $\mathcal{H}_0$ of finite linear combinations of $k(\mathbf{x}, \cdot)$ functions ($\mathbf{x} \in D$),

$$\mathcal{H}_0 = \text{span} (k(\mathbf{x}, \cdot), \mathbf{x} \in D) = \left\{ \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot) : n \geq 1, \mathbf{x}_1, \dots, \mathbf{x}_n \in D, a_1, \dots, a_n \in \mathbb{R} \right\},$$

and endow it with the bilinear form on $\mathcal{H}_0$ defined for $f = \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot)$ and $g = \sum_{j=1}^m b_j k(\mathbf{x}'_j, \cdot)$ ($m \geq 1, \mathbf{x}'_1, \dots, \mathbf{x}'_m \in D, b_1, \dots, b_m \in \mathbb{R}$) by

$$\langle f, g \rangle_{\mathcal{H}_0} = \sum_{i=1}^n \sum_{j=1}^m a_i b_j k(\mathbf{x}_i, \mathbf{x}'_j)$$

Note on the one hand that applying this definition with $g = k(\mathbf{x}, \cdot)$ ($\mathbf{x} \in D$) delivers $\langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}_0} = \sum_{i=1}^n a_i k(\mathbf{x}_i, \mathbf{x}) = f(\mathbf{x})$, a.k.a. the reproducing property. On the other hand, it requires for $\langle f, g \rangle_{\mathcal{H}_0}$ to be well-defined not to depend on the specific expressions of f and g as linear combinations of the $k(\mathbf{x}_i, \cdot)$'s (resp. $k(\mathbf{x}'_j, \cdot)$'s), which is ok indeed since the two following equalities hold by the reproducing property,

$$\begin{aligned} \langle f, g \rangle_{\mathcal{H}_0} &= \sum_{i=1}^n a_i \langle k(\mathbf{x}_i, \cdot), g \rangle_{\mathcal{H}_0} = \sum_{i=1}^n a_i g(\mathbf{x}_i) \\ &= \sum_{j=1}^m b_j \langle f, k(\mathbf{x}'_j, \cdot) \rangle_{\mathcal{H}_0} = \sum_{j=1}^m b_j f(\mathbf{x}'_j). \end{aligned}$$

Besides this, $\langle \cdot, \cdot \rangle_{\mathcal{H}_0}$ is positive semi-definite as, for any $f = \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot)$,

$$\langle f, f \rangle_{\mathcal{H}_0} = \sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) \geq 0,$$

À tout moyen, il existe un espace de Hilbert reproduit (qui est vect($k(x, \cdot)$))

by p.d.-ness of k . Consequently, the Cauchy-Schwarz inequality applies, whereof

$$\forall \mathbf{x} \in D, |f(\mathbf{x})| \leq \sqrt{\langle f, f \rangle_{\mathcal{H}_0}} \sqrt{\langle k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}_0}} = \sqrt{\langle f, f \rangle_{\mathcal{H}_0}} \sqrt{k(\mathbf{x}, \mathbf{x})}.$$

The latter equation proves in turn that $\sqrt{\langle f, f \rangle_{\mathcal{H}_0}} = 0$ implies that $f = 0$ (i.e. $\forall \mathbf{x} \in D, f(\mathbf{x}) = 0$), and so $\langle \cdot, \cdot \rangle_{\mathcal{H}_0}$ does actually constitute an inner product on $\mathcal{H}_0$.

The second and last, slightly more technical bit is to construct the RKHS associated with k from the pre-Hilbert space $(\mathcal{H}_0, \langle \cdot, \cdot \rangle_{\mathcal{H}_0})$. In fact, $(\mathcal{H}_0, \langle \cdot, \cdot \rangle_{\mathcal{H}_0})$ is generally not complete, and so a completion step is required in order to obtain a Hilbert space. Yet, as we know, the completion operation does not necessarily lead to function spaces. Fortunately, the completion of $\mathcal{H}_0$ sketched next will actually deliver a space of functions. Consider indeed a Cauchy sequence $(f_n)_{n \geq 1}$ in $\mathcal{H}_0$. From the reproducing property, we have that at any $\mathbf{x} \in D$ and for any $n, m \geq 1$,

$$|f_n(\mathbf{x}) - f_m(\mathbf{x})| \leq \sqrt{\langle f_n, f_m \rangle_{\mathcal{H}_0}} \sqrt{k(\mathbf{x}, \mathbf{x})}.$$

The Cauchy property of $(f_n)_{n \geq 1}$ in $\mathcal{H}_0$ is thus inherited by the real-valued sequence $(f_n(\mathbf{x}))_{n \geq 1}$, which consequently converges in $\mathbb{R}$. Calling $f(\mathbf{x})$ the obtained limit, we hence construct a function of $\mathbf{x}$ that appears in some sense as limit of $(f_n)_{n \geq 1}$. The remaining endeavors are to show that the space of such limits can be endowed with an inner product consistently constructed from $\langle \cdot, \cdot \rangle_{\mathcal{H}_0}$ (by passing to the limit with sequences of the form $\langle f_n, g_n \rangle_{\mathcal{H}_0}$) and that the resulting inner product space is complete indeed. The latter Hilbert space is the RKHS associated with k . $\square$

We now come back to some of our original motivations of shedding some light on Kriging and related equations from the RKHS viewpoint. Let us now think of $\mathbf{x}_1, \dots, \mathbf{x}_n \in D$ as design points (observation locations), and consider situations where one would like to reconstruct/approximate a function $f \in \mathcal{H}$ based on noiseless or noisy observations of f at these points. Let us now introduce a new notation, namely $S = \text{span}(k(\mathbf{x}_i, \cdot), i = 1, \dots, n)$, and remark that as a finite-dimensional linear subspace of $\mathcal{H}$, S is closed and one can safely define orthogonal projectors π_S and $\pi_{S^\perp}$ on S and its orthogonal in $\mathcal{H}$.

Consider now the case where an approximation $\hat{f}$ of f is sought in S so as to minimize the RKHS distance between $\hat{f}$ and f , leading in fact to $\hat{f} = \pi_S(f)$. In order to calculate this optimal approximation, let us consider an arbitrary $\sum_{i=1}^n \alpha_i k_{\mathbf{x}_i} \in S$ and seek to determine optimal values for the α_i 's. A first step is to note that

$$\begin{aligned} \left\| f - \sum_{i=1}^n \alpha_i k_{\mathbf{x}_i} \right\|_{\mathcal{H}}^2 &= \|f\|_{\mathcal{H}}^2 + \left\| \sum_{i=1}^n \alpha_i k_{\mathbf{x}_i} \right\|_{\mathcal{H}}^2 - 2 \sum_{i=1}^n \alpha_i \langle f, k_{\mathbf{x}_i} \rangle_{\mathcal{H}} \\ &= \|f\|_{\mathcal{H}}^2 + \boldsymbol{\alpha}^T K \boldsymbol{\alpha} - 2 \boldsymbol{\alpha}^T \mathbf{z}, \end{aligned}$$

where $K = (k(\mathbf{x}_i, \mathbf{x}_j))_{i,j \in \{1, \dots, n\}}$ and $\mathbf{z}$ denotes the n -dimensional vector of function evaluations $f(\mathbf{x}_i)$ ($1 \leq i \leq n$). Seen as a function of $\boldsymbol{\alpha}$, the latter is convex and differentiable. Assuming further that K is invertible, this function can be globally minimized by setting its gradient to zero, delivering $K^{-1} \mathbf{z}$ as optimal value for $\boldsymbol{\alpha}$, leading to an optimal approximation of f in $\mathcal{H}$ coinciding with the Simple Kriging mean predictor associated with a centred squared-integrable random field with covariance kernel k . In addition, considering now the projection of $k_{\mathbf{x}}$ onto S , we obtain

by a similar calculation that $\pi_S(k_{\mathbf{x}}) = \sum_{i=1}^n \sum_{j=1}^n k(\mathbf{x}, \mathbf{x}_i) K^{ij} k(\cdot, \mathbf{x}_j) = \mathbf{k}(\mathbf{x}) K^{-1} \mathbf{k}(\cdot)$ (with our notation $\mathbf{k}(\mathbf{x}) = (k(\mathbf{x}, \mathbf{x}_1), \dots, k(\mathbf{x}, \mathbf{x}_n))^T$ from Simple Kriging) and

$$d_{\mathcal{H}}(k_{\mathbf{x}}, S)^2 = \min_{g \in S} \|k_{\mathbf{x}} - g\|_{\mathcal{H}}^2 = \|k_{\mathbf{x}} - \pi_S(k_{\mathbf{x}})\|_{\mathcal{H}}^2 = k(\mathbf{x}, \mathbf{x}) - \mathbf{k}(\mathbf{x}) K^{-1} \mathbf{k}(\mathbf{x}).$$

We will now consider an RKHS function approximation approach dating back —to the best of our knowledge— to Kimeldorf and Wahba (1970), where it is assumed that the observations of f are noisy, or at least that for some reason one wishes to smooth out local variations in the data by appealing to a family of function approximations not necessarily interpolating f at the locations of considered (noisy) evaluations. This is done by minimizing a criterion making a trade-off between fidelity to observed data versus norm of the approximator, in the general spirit of Tikhonov regularization. The following property (particular case of the Kimeldorf-Wahba theorem) illustrates the power of the RKHS framework for function approximation.

Property 2. For any $\lambda > 0$ and $z_1, \dots, z_n$, the criterion

$$Q : g \in \mathcal{H} \rightarrow Q(g) = \frac{1}{n} \sum_{i=1}^n (z_i - g(\mathbf{x}_i))^2 + \lambda \|g\|_{\mathcal{H}}^2 \in [0, \infty),$$

is minimized over $S = \text{span}(k(\mathbf{x}_i, \cdot), i = 1, \dots, n)$. Furthermore, the solution is unique and writes

$$\hat{f}(\mathbf{x}) = \mathbf{k}(\mathbf{x})^T (K + n\lambda I_n)^{-1} \mathbf{z}.$$

Proof. A first, essential step, consists in acknowledging that a minimizer of Q necessarily lies in S . To see this, let us consider any $g \in \mathcal{H}$ and decompose it as the sum of its orthogonal projections onto S and $S^\perp$, i.e. $g = \pi_S(g) + \pi_{S^\perp}(g)$. We will now prove that $Q(\pi_S(g)) \leq Q(g)$ with strict inequality in case $g \notin S$, hence implying that any minimizer of Q necessarily lies in S . On the one hand, by remembering that $g(\mathbf{x}_i) = \langle g, k(\mathbf{x}_i, \cdot) \rangle_{\mathcal{H}}$ and by definition of S and its orthogonal complement, we find that $\langle \pi_{S^\perp}(g), k(\mathbf{x}_i, \cdot) \rangle_{\mathcal{H}} = 0$ whereof $\langle \pi_S(g), k(\mathbf{x}_i, \cdot) \rangle_{\mathcal{H}} = g(\mathbf{x}_i)$ and $\sum_{i=1}^n (z_i - \pi_S(g)(\mathbf{x}_i))^2 = \sum_{i=1}^n (z_i - g(\mathbf{x}_i))^2$. On the other hand, by Pythagoras' theorem, $\|g\|_{\mathcal{H}}^2 = \|\pi_S(g)\|_{\mathcal{H}}^2 + \|\pi_{S^\perp}(g)\|_{\mathcal{H}}^2 \geq \|\pi_S(g)\|_{\mathcal{H}}^2$ with equality for $g \in S$ and strict inequality otherwise, hence proving the first point. As for the unicity of the solution and its explicit form, it follows from the expansion of any candidate solution as $\sum_i \alpha_i k_{\mathbf{x}_i}$ ($\alpha_1, \dots, \alpha_n \in \mathbb{R}$). From there we expand $Q(\sum_i \alpha_i k_{\mathbf{x}_i})$ by noting that

$$\sum_{i=1}^n \left(z_i - \sum_j \alpha_j k_{\mathbf{x}_j}(\mathbf{x}_i) \right)^2 = \|\mathbf{z}\|_n^2 + \boldsymbol{\alpha}^T K^2 \boldsymbol{\alpha} - 2\boldsymbol{\alpha}^T K \mathbf{z}$$

and $\left\| \sum_i \alpha_i k_{\mathbf{x}_i} \right\|_{\mathcal{H}}^2 = \boldsymbol{\alpha}^T K \boldsymbol{\alpha}$

Differentiating the convex polynomial function Q with respect to $\boldsymbol{\alpha}$, we then find $\nabla Q(\boldsymbol{\alpha}) = (\frac{2}{n} K^2 + \lambda K) \boldsymbol{\alpha} - \frac{2}{n} K \mathbf{z}$ and so Q possesses a unique critical point in $(K + n\lambda I_n)^{-1} \mathbf{z}$ (the matrix $K + n\lambda I_n$ being systematically invertible as K is positive semi-definite and $\lambda > 0$), leading to the claimed equation for $\hat{f}$. $\square$

After these considerations in a function approximation framework, let us now come back to random fields and tackle some important connections between centred Gaussian random fields and the RKHSs associated with their respective covariance kernels. As we will see indeed, orthogonal expansions of Gaussian random fields in the flavour of the Karhunen-Loève one have deep connections to deterministic expansions of the associated kernels.

The key to this connections lies in a Hilbert space isometric isomorphism, commonly referred to as “the Loève Isomorphism”, that bi-uniquely links random variables from the Hilbert space generated by a centred Gaussian random field possessing a covariance kernel k with deterministic functions from the corresponding RKHS. Before we present this isomorphism in more detail, we need to say a few more words about the notion of Hilbert space generated by a (centred) Gaussian random field.

Let us denote by $Z = (Z_{\mathbf{x}})_{\mathbf{x} \in D}$ a centred Gaussian random field with covariance kernel k , and by $\mathcal{L}_0 = \{\sum_{i=1}^n a_i Z_{\mathbf{x}_i}, n \geq 1, a_1, \dots, a_n \in \mathbb{R}, \mathbf{x}_1, \dots, \mathbf{x}_n \in D\}$ the set of random variables writing as finite linear combinations of pointwise evaluations of Z . By Gaussianity of Z , the elements of $\mathcal{L}_0$ are all Gaussian (and any random vector with components in $\mathcal{L}_0$ forms a Gaussian vector), so in particular they are square-integrable. Endowed with the $L^2(\mathbb{P})$ scalar product (where $\mathbb{P}$ is the probability measure of the probability triplet underlying Z), $\mathcal{L}_0$ hence constitutes an inner product space. Yet it is generally not complete, and what is called *the Hilbert space generated by Z* is the completion of $\mathcal{L}_0$ with respect to the $L^2(\mathbb{P})$ norm:

$$\mathcal{L} = \overline{\mathcal{L}_0}^{\|\cdot\|_{L^2(\mathbb{P})}} = \overline{\text{span}(Z_{\mathbf{x}}, \mathbf{x} \in D)}^{\|\cdot\|_{L^2(\mathbb{P})}}, \quad (1)$$

which hence occurs to be a Hilbert space of Gaussian random variables, as its elements are by construction $L^2(\mathbb{P})$ limits of sequences of Gaussian random variables. We are now ready to introduce the *Loève* (or *canonical*) *isomorphism* between $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ and $(\mathcal{L}, \langle \cdot, \cdot \rangle_{\mathcal{L}})$, where $\langle \cdot, \cdot \rangle_{\mathcal{L}}$ denotes the restriction of $\langle \cdot, \cdot \rangle_{L^2(\mathbb{P})}$ to $\mathcal{L}$. We now follow the exposition made of it in Chapter 3 of Adler and Taylor’s *Geometry of Random Fields* (2007), and denote for a start this isomorphism by Θ . $\Theta : \mathcal{H} \rightarrow \mathcal{L}$ is essentially defined by its values at $k(\mathbf{x}, \cdot)$ ($\mathbf{x} \in D$):

$$\Theta(k(\mathbf{x}, \cdot)) = Z_{\mathbf{x}}, \quad (2)$$

as from there it can be straightforwardly extended to linear combinations:

$$\Theta\left(\sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot)\right) = \sum_{i=1}^n a_i Z_{\mathbf{x}_i}, \quad (3)$$

where $n \geq 1, a_1, \dots, a_n \in \mathbb{R}, \mathbf{x}_1, \dots, \mathbf{x}_n \in D$. At this stage it is useful to remark that Θ is norm-preserving on $\mathcal{L}_0$, in the sense that for the latter arbitrary setting of points and weights,

$$\begin{aligned} \left\| \sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot) \right\|_{\mathcal{H}}^2 &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j k(\mathbf{x}_i, \mathbf{x}_j) \\ &= \left\| \sum_{i=1}^n a_i Z_{\mathbf{x}_i} \right\|_{\mathcal{L}}^2 = \left\| \Theta\left(\sum_{i=1}^n a_i k(\mathbf{x}_i, \cdot)\right) \right\|_{\mathcal{L}}^2, \end{aligned} \quad (4)$$

and passing to the limit in Eqs.(3)-(4) delivers the sought isometric isomorphism between $\mathcal{H}$ and $\mathcal{L}$. As we will discuss now, this paves the way to very useful representations for Z , notably in terms of orthonormal bases of $\mathcal{H}$ when they do exist.

We assume here for simplicity that D is a compact metric space and k is continuous on $D \times D$, implying that $\mathcal{H}$ is separable. As a separable Hilbert space, $\mathcal{H}$ possesses a countable basis, say $(\psi_j)_{j \geq 1}$ (our notation slightly differs from the one of Adler and Taylor 2007). Setting

$$\varepsilon_j = \Theta(\psi_j) \quad (j \geq 1), \quad (5)$$

we then obtain a countable basis, i.e. $(\varepsilon_j)_{j \geq 1}$, for the space $\mathcal{L}$. In particular, the ε_j are necessarily $\mathcal{N}(0, 1)$ -distributed and independent, and for any $\mathbf{x} \in D$,

$$Z_{\mathbf{x}} = \sum_{j=1}^{\infty} \varepsilon_j \langle Z_{\mathbf{x}}, \varepsilon_j \rangle_{L^2(\mathbb{P})} = \sum_{j=1}^{\infty} \varepsilon_j \mathbb{E}[Z_{\mathbf{x}} \varepsilon_j] \quad (6)$$

where the convergence is in $L^2(\mathbb{P})$. Now, remembering the isometry property of Θ and the reproducing property in $\mathcal{H}$, we get the following reformulation

$$\mathbb{E}[Z_{\mathbf{x}} \varepsilon_j] = \langle k(\mathbf{x}, \cdot), \psi_j \rangle_{\mathcal{H}} = \psi_j(\mathbf{x}) \quad (j \geq 1). \quad (7)$$

To summarize, for any orthonormal basis $(\psi_j)_{j \geq 1}$ for $\mathcal{H}$,

$$Z_{\mathbf{x}} = \sum_{j=1}^{\infty} \varepsilon_j \psi_j(\mathbf{x}), \quad (8)$$

where $(\varepsilon_j)_{j \geq 1}$ is the orthonormal sequence of Gaussian random variables given by $\varepsilon_j = \Theta(\psi_j)$ ($j \geq 1$). Let us stress however that, although at any fixed (but arbitrary) $\mathbf{x} \in D$, the convergence of the series in Eqs. 6 and 8 is in the $L^2(\mathbb{P})$ sense. Yet, the following theorem provides a useful complement at the price of assuming a.s. continuity for Z :

Theorem 1. *If Z is almost surely continuous, then the sum in Eq. 8 converges uniformly on D with probability 1.*

Proving this theorem can be done based on the two following lemmas. The proof of the first lemma is left as an exercise, while we accept here the second one, a convergence result attributed to Lévy in the scalar case and to Itô and Nisio in the presented more general framework.

Lemma 1. *Let $(V_j)_{j \geq 1}$ be a sequence of symmetric independent random variables taking values in a separable, real Banach space B equipped with the norm topology. Let $S_n = \sum_{j=1}^n V_j$. Then S_n converges with probability one if and only if there exists a B -valued random variable S such that $\langle S_n, x^* \rangle \rightarrow \langle S, x^* \rangle$ in probability for any $x^* \in B^*$, where B^* is the topological dual of B .*

Lemma 2. *Let $(\psi_j)_{j \geq 1}$ be an orthonormal basis for $\mathcal{H}$. Then, under the continuity assumption of k , each ψ_j is continuous and $\sum_{j=1}^n \psi_j(\mathbf{x})^2$ converges uniformly in $\mathbf{x} \in D$ to $k(\mathbf{x}, \mathbf{x})$.*

Proof of Theorem 1. For any $\mathbf{x} \in D$, $\sum_j^n \varepsilon_j \psi_j(\mathbf{x})$ is a sum of independent random variables converging in $L^2(\mathbb{P})$. By Lemma 1 applied in the real-valued case, $\sum_j^n \varepsilon_j \psi_j(\mathbf{x})$ converges with probability 1 to $Z_{\mathbf{x}}$. Now the random field Z , assumed to have continuous paths almost surely, can be seen as a random element in the Banach space $C(D)$ of continuous functions on D equipped with the sup-norm topology, as each $\varepsilon_j \psi_j$ can also be seen. Defining $Z^{(n)}(\cdot) = \sum_j^n \varepsilon_j \psi_j(\cdot) = \sum_j^n \Theta(\psi_j) \psi_j(\cdot)$ and recalling that the topological dual of $C(D)$ is made of finite, signed Borel measures μ on D with $\langle f, \mu \rangle = \int_D f d\mu$ (for $f \in C(D)$), it suffices to show by Lemma 1 that for every $\mu \in C(D)^*$, the sequence of random variables $\langle Z^{(n)}, \mu \rangle$ converges in probability to $\langle Z, \mu \rangle$. Yet,

$$\begin{aligned} \mathbb{E} [|\langle Z^{(n)}, \mu \rangle - \langle Z, \mu \rangle|] &= \mathbb{E} \left[\left| \int_D (Z_{\mathbf{x}}^{(n)} - Z_{\mathbf{x}}) d\mu(\mathbf{x}) \right| \right] \\ &\leq \int_D \mathbb{E} [|Z_{\mathbf{x}}^{(n)} - Z_{\mathbf{x}}|] d|\mu|(\mathbf{x}) \\ &\leq \int_D \mathbb{E} [|Z_{\mathbf{x}}^{(n)} - Z_{\mathbf{x}}|^2]^{\frac{1}{2}} d|\mu|(\mathbf{x}) \\ &= \int_D \left(\sum_{j=n+1}^{\infty} \psi_j(\mathbf{x})^2 \right)^{\frac{1}{2}} d|\mu|(\mathbf{x}), \end{aligned}$$

where $|\mu|$ is the total variation measure for μ . Finally, since $\sum_{j=n+1}^{\infty} \psi_j(\mathbf{x})^2 \rightarrow 0$ uniformly in $\mathbf{x} \in D$ by Lemma 2, the last upper bound tends to 0 as $n \rightarrow \infty$. Since this implies convergence in probability of $\langle Z^{(n)}, \mu \rangle$ to $\langle Z, \mu \rangle$, the proof is complete. $\square$

Let us now conclude this excursion on RKHSs by briefly reviewing the Karhunen-Loève (KL) expansion of Gaussian random fields (here under the assumptions that $D = [0, 1]^d$ and k is continuous), in light of our previous considerations about the Loève isomorphism. Here we consider the Lebesgue measure on D and stick to the colloquial notation $d\mathbf{x}$ for simplicity. The starting point of KL is to perform the spectral decomposition of the operator

$$T_k : \phi \in L^2(D) \rightarrow T_k(\phi)(\cdot) = \int_D k(\mathbf{x}, \cdot) \phi(\mathbf{x}) d\mathbf{x} \in L^2(D). \quad \leftarrow$$

By classical spectral theory, it is known that this self-adjoint compact operator possesses non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ and respectively associated normalized eigenfunctions $\varphi_1, \varphi_2, \dots \in L^2(D)$ forming an orthonormal basis of $L^2(D)$. The following theorem gives an expansion of k in terms of λ_j 's and φ_j 's:

Theorem 2. (Mercer) For k , $(\lambda_j)_{j \geq 1}$, $(\varphi_j)_{j \geq 1}$ as above,

$$k(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^{\infty} \lambda_j \varphi_j(\mathbf{x}) \varphi_j(\mathbf{x}'), \quad ? \quad \text{réduction de rang? (cf Salem et Sauter)?}$$

where the series converges absolutely and uniformly on $D \times D$.

The Karhunen-Loève expansion is then obtained by relying on the previously reviewed results on the Loève isometry in the case where $\psi_j = \sqrt{\lambda_j} \varphi_j$ ($j \geq 1$). That the latter define indeed an orthonormal basis of $\mathcal{H}$ is left as yet another exercise.

domaine compact